

WU SU

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EDUCATION

2021 - Present **Center for Data Science, Academy for Advanced Interdisciplinary Studies** **Peking University, China**
Ph.D. candidate in Data Science (expected 2026), supervised by Prof. Song Xi Chen

2016 - 2020 **College of Sciences** **China Jiliang University, China**
B.S. in Information and Computing Science

RESEARCH INTERESTS

My research interests include *data fusion and data assimilation*, *high-dimensional statistics*, and *physics-informed statistical learning*, motivated by applications in Earth and ocean sciences.

- **Data fusion and data assimilation:** Developing statistically principled methods for integrating heterogeneous observations into dynamical Earth and ocean systems, with an emphasis on building scalable data assimilation systems for large-scale, real-world deployment.
- **Physics-informed statistical learning:** Leveraging prior physical knowledge to guide statistical modeling by incorporating physics-based constraints (e.g., via loss/regularization), while using stochastic-process models to represent and quantify the uncertain components of physical processes.
- **High-dimensional statistics under computational constraints:** Investigating the optimality of high-dimensional ($n \ll p$) methods under computational constraints, particularly for large-scale data with latent structure, such as those arising in Earth sciences.

My future research aims to develop deployable methods for Earth-system modeling, centered on data assimilation and real-time decision-making, to support ocean prediction, climate analysis, and carbon-flux estimation, and to advance Digital Twin Earth.

PROJECTS AND INTERDISCIPLINARY EXPERIENCE

Multi-sphere Interaction Data Integration in Western Pacific Ocean 2023–Present

- Development of a North Pacific ocean data assimilation and forecasting system:
 - Integrated high-dimensional statistical methods into ocean data assimilation using blockwise, multigrid, and spatially localized weighting strategies, combined with physical constraints from Navier–Stokes equations, overcoming computational and theoretical bottlenecks in high-resolution assimilation.
 - Achieved assimilation skill superior to state-of-the-art benchmark from Mercator Ocean International.
 - Established an end-to-end operational workflow (observation quality control, assimilation, forecast, evaluation), now stably running on the Laoshan National Laboratory supercomputing platform.
- Design and execution of a targeted eddy observation experiment in the Western Pacific:
 - Developed real-time ocean current estimation and data-driven experimental design with adaptive control of autonomous platforms, enabling tight coupling between observation and prediction.
 - Achieved the first fine-scale networked observation, to our knowledge, of a mesoscale eddy.

High-resolution Ensemble Kalman Filter and Carbon Flux Fused Algorithm 2022–Present

- Development of a global carbon-flux data assimilation system:
 - Proposed a multi-inflation strategy to jointly assimilate satellite and in situ CO₂ observations, adaptively accounting for observational heterogeneity and improving inversion stability and accuracy beyond existing benchmark products (e.g., CarbonTracker).
 - Coupled with the global atmospheric chemistry transport model GEOS-Chem to perform high-resolution global carbon-flux inversion and quantify terrestrial and oceanic carbon sinks.
 - Provided systematic uncertainty diagnostics to support carbon budget accounting and emission assessment; this work received a nomination for the Eni Award.

PUBLICATIONS AND PAPERS UNDER REVIEW

- **Su, W.**, Li, H., Jing, Z., Qiu, Y. and Chen, S.X. (2026+). Physics-informed Statistical Data Fusion for Reconstructing 3D Current Fields of Oceanic Eddies. Major revision for *Journal of the American Statistical Association*.
- **Su, W.** and Qiu, Y. (2026+). High-dimensional Clustering and Signal Recovery under Block Signals. Moderate revision for *Journal of the American Statistical Association*. [\[arxiv\]](#)
- Qian, L., **Su, W.**, Huang, Y., and Chen, S.X. (2026+). Likelihood Matching for Diffusion Models. Under review. [\[arxiv\]](#)
- **Su, W.**, E, X., Jing, Z. and Chen, S.X. (2026). Glider Path Design and Control for Reconstructing Three-Dimensional Structures of Oceanic Mesoscale Eddies. *Proceedings of the Royal Society A*, 482(2333):20250533.[\[link\]](#)
- Du, T., Zhang, Y., **Su, W.**, Jing, Z., Chen, Z., Qiu, Y., Zhang, H., Yang, S. and Ma, W. (2025). Three-Dimensional Observations of a Mesoscale Eddy in the Kuroshio Extension Based on Multiple Platforms. *Scientific Data*. 12: 2013. [\[link\]](#)[\[pdf\]](#)
- Weng, Z., Zhu, L., Li, J., Zhang, Y., Liu, X., **Su, W.**, Sun, H., Li, X. (2025). Using EnKF Data Assimilation to Improve Predictions of Volcanic Ash Dispersion. *Journal of Geophysical Research: Atmospheres*, 130:e2024JD042215. [\[link\]](#)[\[pdf\]](#)
- **Su, W.**, Wang, B., Chen, H., Zhu, L., Zheng, X. and Chen, S.X. (2024). A new global carbon flux estimation methodology by assimilation of both in situ and satellite CO₂ observations. *npj Climate and Atmospheric Science*, 7:287. [\[link\]](#)[\[pdf\]](#)
- Tong, P., **Su, W.**, Li, H., Ding, J., Zhan, H. and Chen, S.X. (2023). Distribution Free Domain Generalization. *Proceedings of the 40th International Conference on Machine Learning (ICML 2023)*, 202:34369-34378. [\[link\]](#)[\[pdf\]](#)

INVITED TALKS

- **Minimax Optimal Clustering under Block Signals and Computational Constraint**
 - 2025/07 The third Joint Conference on Statistics and Data Science in China (Hangzhou, China)
- **Glider Path Design and Control for Surveying an Oceanic Eddy**
 - 2024/11 The 12th National Probability and Statistics Conference of China (Xiamen, China)
- **Multi-source Observation Global Carbon Flux Estimation using EnKF**
 - 2024/07 The second Joint Conference on Statistics and Data Science in China (Kunming, China)
 - 2025/04 *Invited seminar*, National Satellite Meteorological Center (Beijing, China).